MIL-D-20305
21 November 1951

MILITARY SPECIFICATION

DIMETHYLANILINE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-11-30C
17 June 1943

This specification consists of this cover sheet and Specification 50-11-30C, dated 17 June 1943, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
Army-O

GPO 83-40257

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U. S. ARMY
SPECIFICATION

No. 50-11-30C
June 17, 1943
Superseding
No. 50-11-30B
October 5, 1938

DIMETHYLANILINE

A. APPLICABLE SPECIFICATIONS

A-1. The following U.S.Army Specifications, of the issues in effect on the date of invitation for bids, form a part of this specification:

No. 50-0-1, General Specification for Ammunition Except Small Arms Ammunition
No. 100-2, Standard Specifications for Marking Shipments.

B. GRADE.

B-1. This specification covers one grade of dimethylaniline.

C. MATERIALS AND WORKMANSHIP.

C-1. There are no requirements relative to materials or workmanship applicable to this specification.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. Color.- Yellow
E-2. Freezing point. - Minimum, 1.9° C.
E-3. Residue. - Maximum, 0.1 percent.
E-4. Solubility. - 100 percent.
E-5. Alkalinity or acidity. - Maximum, 0.005 percent.
E-6. Aniline. - Maximum, 0.1 percent.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection. Inspection shall be in accordance with Specification No. 50-0-1 and shall be at the point of delivery unless otherwise specified in the contract.

F-2. Size of lots. Maximum 30,000 pounds, if shipment is made in containers other than tank cars. If shipment is made in tank cars, a lot shall consist of one tank car load.

F-3. Sampling shall be done as follows: If the dimethylaniline is shipped in drums or smaller containers, select 10 percent but not more than 10 of the containers so as to be representative of the lot. Mix the contents of each selected container, remove approximately 1 quart of material and place this in a tightly stoppered bottle. Label each primary sample so

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that the container from which it was taken can be identified. Make a composite sample of approximately 1 quart, comprising equal parts of each primary sample, and mix this thoroughly. Place the composite sample in a tightly stoppered bottle. If the material is shipped in a tank car, remove the sample (approximately 1 quart) by means of a thief or a clean weighted bottle. Rinse the bottle, which has a capacity of approximately 1 quart, with the material being sampled. Lower the unstoppered bottle by means of a cord, chain, or rod to the bottom of the tank, and immediately withdraw to the surface. In order to obtain a representative cross section sample of the material, raise and lower the bottle at a uniform rate of speed, and regulate the speed so that the bottle is just filled as it reaches the surface of the liquid. Place this sample in a tightly stoppered bottle. Label the sample so as to show the name of the material, manufacturer, plant, purchase order, and number of pounds in the lot. Make all acceptance tests on the sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements.

F-4. Laboratory examination shall be made as follows:

F-4a. Color. - Determine by visual inspection.

F-4b. Freezing point. - Dry a portion of approximately 50 cc. of the sample by adding 1 cc. of absolute alcohol, heating the solution for 1 hour in an oven at 95° to 100° C., and cooling in a sulphuric acid vacuum desiccator. Place the dried dimethylaniline in the test tube of the freezing point apparatus shown in Figure 1. Insert the aluminum rod and a standardized total immersion thermometer, graduated in 0.2° C. or smaller subdivisions and extending over the range -10° to + 10° C., into the cork stopper. Fasten a correction thermometer adjacent to the standardized thermometer so that its bulb will be approximately level with the center of the exposed mercury column when the dimethylaniline freezes. Insert the cork assembly into the test tube, and adjust the standardized thermometer so that its bulb is at least 1/2 inch from the bottom of the test tube. Clamp the tube assembly in the ice water bath so that the level of the dimethylaniline is at least 1 inch below the level of the bath. Stir the sample throughout the determination, making approximately 120 complete up and down motions of the aluminum rod per minute. Watch the thermometer closely to determine when the temperature of the sample ceases falling and begins to rise due to the heat of crystallization. Note the temperature of the sample every 15 seconds until the maximum is reached, and record this temperature as the uncorrected freezing point. Note the reading on the correction thermometer and consider this value as the average temperature of the exposed mercury column. Calculate the correction for emergent stem as follows:

$$\text{Emergent stem correction} = n\,(T - t) \times 0.000159$$

where n = number of degrees in the exposed mercury column.
T = uncorrected freezing point.
t = average temperature of the exposed mercury column.

Calculate the freezing point of the sample by adding the emergent stem and standardization corrections to the uncorrected freezing point value. For samples containing more than 1 percent of impurities, add salt to the bath if the dimethylaniline does not freeze in the ice water bath.

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2 - 250 cc. filter funnels

F-4c. Residue. Transfer a weighed portion of approximately 100 grams of the sample to a tared evaporating dish. By means of a hot plate evaporate the liquid rapidly to dryness. Cool the dish and contents in a desiccator and weigh. Calculate the increase in weight to percentage of residue in the sample.

F-4d. Solubility. Add a 1 cc. portion of the sample to 10 cc. of 94 to 96 percent sulphuric acid solution and shake the mixture. Note if the dimethylaniline dissolves completely in the acid solution. Pour the solution into 50 cc. of distilled water. Note if there is any separation of an oil or solid matter, indicating that less than 100 percent of the dimethylaniline is soluble in sulphuric acid solution.

F-4e. Alkalinity or acidity. Place 20 cc. of neutral distilled water in a 100 cc. separatory funnel and add a 25 gram portion of the sample. Shake the mixture thoroughly for 2 minutes and allow it to stand for a few minutes to permit separation into two layers. Draw off the lower aqueous layer and by means of a pipette transfer 10 cc. of this aqueous layer to a 100 cc. Erlenmeyer flask. Add 3 drops of rosolic acid indicator and titrate the solution with approximately N/10 hydrochloric acid or sodium hydroxide solutic Calculate the percentage of alkalinity or acidity of the sample.

$$\text{Percentage of alkalinity, as sodium hydroxide} = \frac{8.001\ AB}{C}$$

$$\text{Percentage of acidity, as hydrochloric acid} = \frac{7.293\ AB}{C}$$

where A = volume of acid or alkali solution used.
B = normality of acid or alkali solution.
C = weight of sample.

F-4f. Aniline. By means of a pipette transfer 5 cc. portions of the sample to each of 2 glass tubes. To each tube add 15 cc. of ether and 5 cc. of calcium oxychloride solution. Prepare the calcium oxychloride solution by shaking 200 cc. of distilled water with not less than 35 grams of calcium hypochlorite, containing not less than 30 percent of available chlorine, until no further solution takes place. Allow the mixture to settle and filter the solution. The tubes should be such that the two 25 cc. volumes of liquid form columns of equal height. Stopper the tubes, shake thoroughly, and allow to stand for 30 to 35 minutes. Compare the reddish colors produced in the ether layers with those produced in blanks started at the same time with freshly distilled pure dimethylaniline to which 0.1 percent of aniline has been added. Note if the colors produced in the samples are no darker than those produced in the blanks, indicating the presence of not more than 0.1 percent of aniline.

G. PACKING AND MARKING.

G-1. Packing. Unless otherwise specified, dimethylaniline shall be delivered in standard commercial containers such as steel drums or tank cars, so constructed and protected as to conform with applicable regulations of the Interstate Commerce Commission and insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery

G-2. Marking. In accordance with U. S. Army Specification No. 100-2

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and applicable regulations of the Interstate Commerce Commission.

H. NOTES.

H-1. Uses. Dimethylaniline covered by this specification is intended for use in the manufacture of tetryl.

H-2. The use of this specification whenever applicable is mandatory on all procuring agencies of the Army.

H-3. If the composite sample representative of the lot fails to comply with the requirements, the manufacturer shall have the option of having analysis made of each primary sample at his own expense. The manufacturer may then replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and again submit the lot for inspection and acceptance, providing that the markings on the containers are such that complete replacement of defective portions of the lot can be made to the satisfaction of the inspector.

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NOTE: Copies of this specification may be obtained from the Office, Chief of Ordnance, War Department, Washington, D. C.

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2. Acetic acid: 5 percent. Prepared by dilution

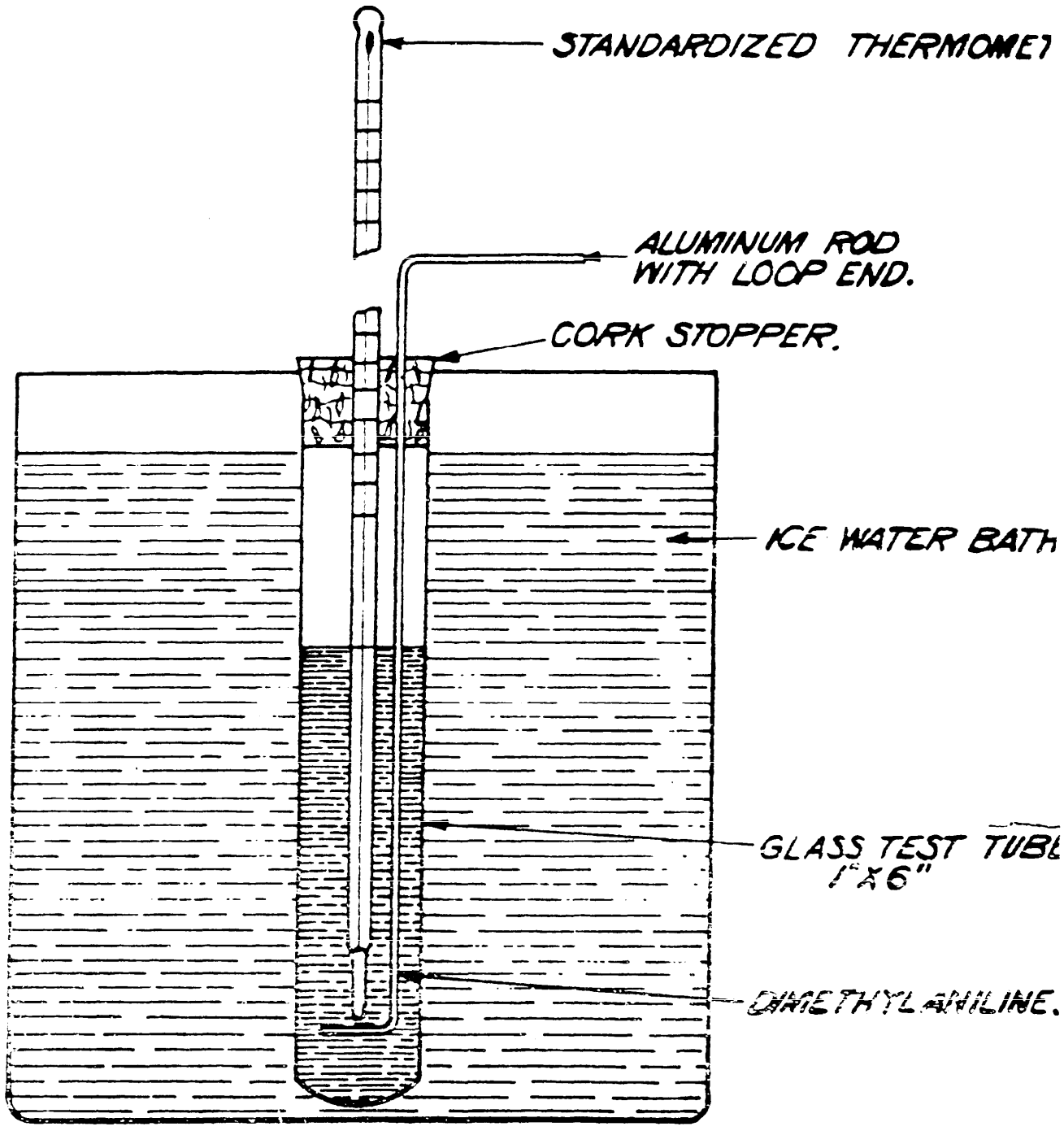


FIGURE I

FREEZING POINT APPARATUS

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OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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